

Fig. 6: An actual-size, single-side PCB for the RF receiver

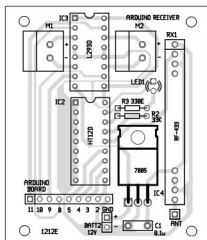


Fig. 7: Component layout for the PCB in Fig. 6

are identical, when any of the switches on the transmitter (marked as FWD, REV, RIGHT, LEFT) is pressed, the corresponding data pin of the decoder

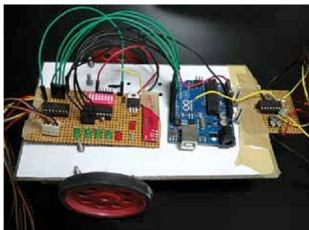


Fig. 8: Author's prototype

goes low. The data outputs from D8 through D11 of HT12D (IC2) are fed to pins 2 through 5 of Arduino UNO board to generate appropriate logic outputs from pins 8 through 11 of Arduino UNO board.

Outputs from pins 8 through 11 of Arduino Uno board are fed to IN1 through IN4 of L293D (IC3) to drive both the motors M1 and M2 as shown in Fig. 3. Outputs OUT1 and OUT2 drive motor M1, and outputs OUT3 and OUT4 drive motor M2. Enable pins EN1 (pin 1) and EN2 (pin 9) are connected to Vcc for always enabled output. Regulator 7805 (IC4) provides regulated 5V to the receiver.

Construction

An actual-size, single-side PCB for the RF transmitter (Fig. 2) is shown in Fig. 4 and its component layout in Fig. 5. The

PCB for the receiver (Fig. 3) is shown in Fig. 6 and its component layout in Fig. 7. A suitable connector arrangement has been made on the RF receiver PCB in order to extend connections to the drive motors and the battery mounted on the chassis of the RF robot.

Software

The source code file (RFROBOT.INO) for

this project is listed at the end of this article. The Arduino Uno is programmed with Arduino IDE software. The ATmega328 on Arduino Uno comes pre-burned with a bootloader that allows you to upload new code to it without using an external hardware programmer. It communicates using the original STK500 protocol. You can also bypass the bootloader and program the microcontroller through the ICSP (in-circuit serial programming) header but using the bootloader makes the programming quick and easy.

Select Arduino Uno from the Tools→Board menu (according to the microcontroller on your board) in the Arduino IDE and burn the program through the standard USB port in the computer.

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RFROBOT.INO

```
//RF Robot
int sw1 = 2;
int sw2 = 3;
int sw3 = 4;
int sw4 = 5;
int out1 = 8;
int out2 = 9;
int out3 = 10;
int out4 = 11;
void setup()
{
  pinMode(sw1, INPUT);
  pinMode(sw2, INPUT);
  pinMode(sw3, INPUT);
  pinMode(sw4, INPUT);
  pinMode(out1, OUTPUT);
  pinMode(out2, OUTPUT);
  pinMode(out3, OUTPUT);
  pinMode(out4, OUTPUT);
}
void loop()
{
  if ((digitalRead(sw1) == LOW)
  && (digitalRead(sw2) == HIGH)
  && (digitalRead(sw3) == HIGH)
  && (digitalRead(sw4) == HIGH))
  {
    fwd();
  }
  else if ((digitalRead(sw1) == HIGH)
  && (digitalRead(sw2) == LOW)
  && (digitalRead(sw3) == HIGH)
  && (digitalRead(sw4) == HIGH))
  {
    bkw();
  }
  else if ((digitalRead(sw1) == HIGH)
  && (digitalRead(sw2) == HIGH)
  && (digitalRead(sw3) == LOW)
  && (digitalRead(sw4) == HIGH))
  {
    lft();
  }
  else if ((digitalRead(sw1) == HIGH)
  && (digitalRead(sw2) == HIGH)
  && (digitalRead(sw3) == HIGH)
  && (digitalRead(sw4) == LOW))
  {
    rgt();
  }
  else
  {
    digitalWrite(out1, LOW);
    digitalWrite(out2, LOW);
    digitalWrite(out3, LOW);
    digitalWrite(out4, LOW);
  }
}

void fwd()
{
  digitalWrite(out1, HIGH);
  digitalWrite(out2, LOW);
  digitalWrite(out3, HIGH);
  digitalWrite(out4, LOW);
}

void bkw()
{
  digitalWrite(out1, LOW);
  digitalWrite(out2, HIGH);
  digitalWrite(out3, LOW);
  digitalWrite(out4, HIGH);
}

void lft()
{
  digitalWrite(out1, LOW);
  digitalWrite(out2, HIGH);
  digitalWrite(out3, HIGH);
  digitalWrite(out4, LOW);
}

void rgt()
{
  digitalWrite(out1, HIGH);
  digitalWrite(out2, LOW);
  digitalWrite(out3, LOW);
  digitalWrite(out4, HIGH);
}
```